

Gym Management System

PRESENTATION • IT 3003

Gym Management System

Enterprise Fitness Operations • Spring Boot & React Pages Application

Course: Advance Programming- IT 3003

Team Name: Team 09(UOC)

Team Members & Responsibilities:

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Project Introduction & Problem Statement

CURRENT INDUSTRY PAIN POINTS

- Manual Paper Logbooks: Error-prone attendance tracking and misplaced records.
- Fragmented Billing: Subscriptions tracked in spreadsheets leading to lost revenue.
- Maintenance Delays: Equipment breakdowns go unnotified until members complain.
- Staff Disconnect: No unified roster management for trainer schedules or salaries.
- Zero Analytics: Lack of real-time KPIs on revenue or gym peak access hours.

PROPOSED SYSTEM OBJECTIVES

- Centralized CRUD Operations: Full lifecycle tracking for Members, Staff, & Equipment.
- Sub-second QR Check-ins: Instant barcode verification with status flags.
- Automated Invoicing: Streamlined subscription plans (VIP, Standard, Basic).
- Recharts KPI Analytics: Live revenue metrics and peak attendance distribution.
- Dark Luxury Ergonomics: Eye-safe UI built with Tailwind CSS & micro-animations.

Technology Stack & Tool Rationale

Architecture Layer	Selected Technology	Technical Rationale
Backend Language	Java 17	Robust object-oriented paradigm, type safety, enterprise reliability, and vast ecosystem.
Backend Framework	Spring Boot 3.x (REST)	Rapid micro-framework setup, embedded Tomcat, seamless Spring Security & Spring Data JPA integration.
Database	H2	H2 gives the project a frictionless development experience, rapid feedback loops, and a clean path to upgrade to a persistent relational database when the application graduates to production.
Frontend	React 19 (Vite), Tailwind CSS, Lucide React	Reactive virtual DOM, sub-millisecond Vite build times, utility-first dark styling, accessible icons.
IDE	IntelliJ IDEA	Industry standard for Java/Spring development, auto-refactoring, bytecode analyzer, database tool window.
Version Control	Git + GitHub	Distributed version control, feature branch workflow, peer code reviews, CI workflow hooks.
Build Tools	Maven (Java), npm / Vite (Frontend)	Dependency lifecycle automation, deterministic lockfiles, rapid HMR during development.
Testing	JUnit 5, Mockito, React Testing Library	Unit tests for Spring business logic, mock repositories, component integration testing.

System Architecture & Data Flow (C4 Model)

LAYER 01: Client / Admin Web Browser

Renders dark-themed React single page app.

Technology: Chrome / Firefox SPA

LAYER 02: REST API Controller Layer

Validates session tokens and exposes REST endpoints.

Technology: Spring Boot 3

LAYER 03: Service & JPA Repository

Executes core business rules and JPA entity queries.

Technology: Java 17 @Service & Hibernate

LAYER 04: Relational Database

Persists Members, Trainers, Equipment, & Payments.

Technology: H2

System Features & Module Breakdown

Member Management

Comprehensive CRUD operations for gym member lifecycle tracking.

- Register new members with tiered plans (VIP, Standard, Basic)
- Edit member details & contact information
- Deactivate / Soft-delete expired memberships
- Instant real-time search by name, email, or plan
- Server-side & UI pagination controls

Trainer Management

Manage staff schedules, specializations, and salary tiers.

- Full CRUD management for fitness instructors
- Specialization categorization (CrossFit, Strength, Yoga, Cardio)
- Salary tracking & active status configuration
- Member-to-trainer ratio monitoring

Equipment Inventory

Preventative maintenance logging and gym asset tracking.

- Track fitness equipment quantity & condition (Optimal, Maintenance, Out of Service)
- Automated routine maintenance schedule alerts
- Equipment categorization (Cardio, Free Weights, Machines)
- Low-stock and repair status flags

Attendance Tracking

Fast, contact-free check-in logging with timestamp verification.

- Real-time check-in via QR / RFID barcode simulation
- Daily member & trainer attendance logs
- Automatic membership expiry warning during check-in
- Peak hours access telemetry graphs

Payments & Revenue

Centralized financial record-keeping and membership revenue analytics.

- Automated invoice generation for plan renewals
- Integrated financial dashboard & monthly revenue charts
- Payment status tracking (Paid, Pending, Overdue)
- Flexible payment method logging (Credit Card, Bank, Cash)

Admin Dashboard & Security

Executive control hub for administrators with secure session authorization.

- Centralized real-time KPI overview cards
- Simple admin-only authentication using session tokens
- Dark-themed UI with ambient gold accent micro-animations
- System REST API response time & query log inspector

Overall System Activity & Workflow

STEP 01

Admin Authentication

Verify session token & authorize admin access.

STEP 02

Navigation & Module Select

Select Members, Trainers, Equipment, or Payments.

STEP 03

Form Input & Validation

Front-end state validation (email format, phone, salary).

STEP 04

REST API Execution

HTTP POST/PUT/DELETE payload sent to Spring Boot.

STEP 05

JPA Persistence

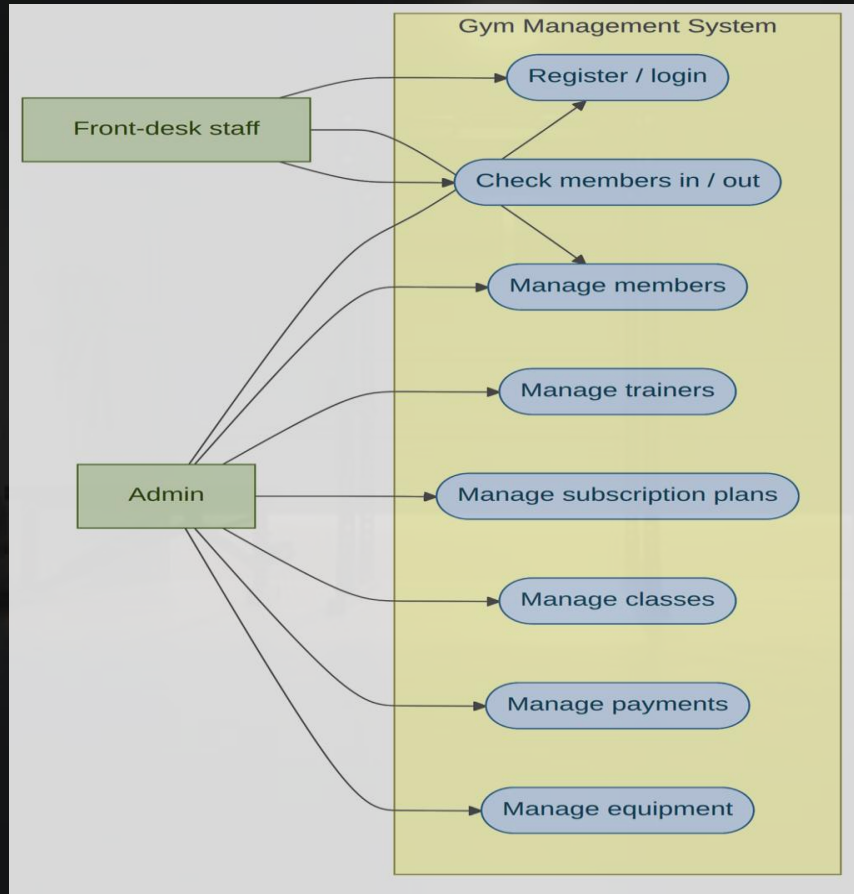
Spring Data JPA executes SQL transaction on MySQL.

STEP 06

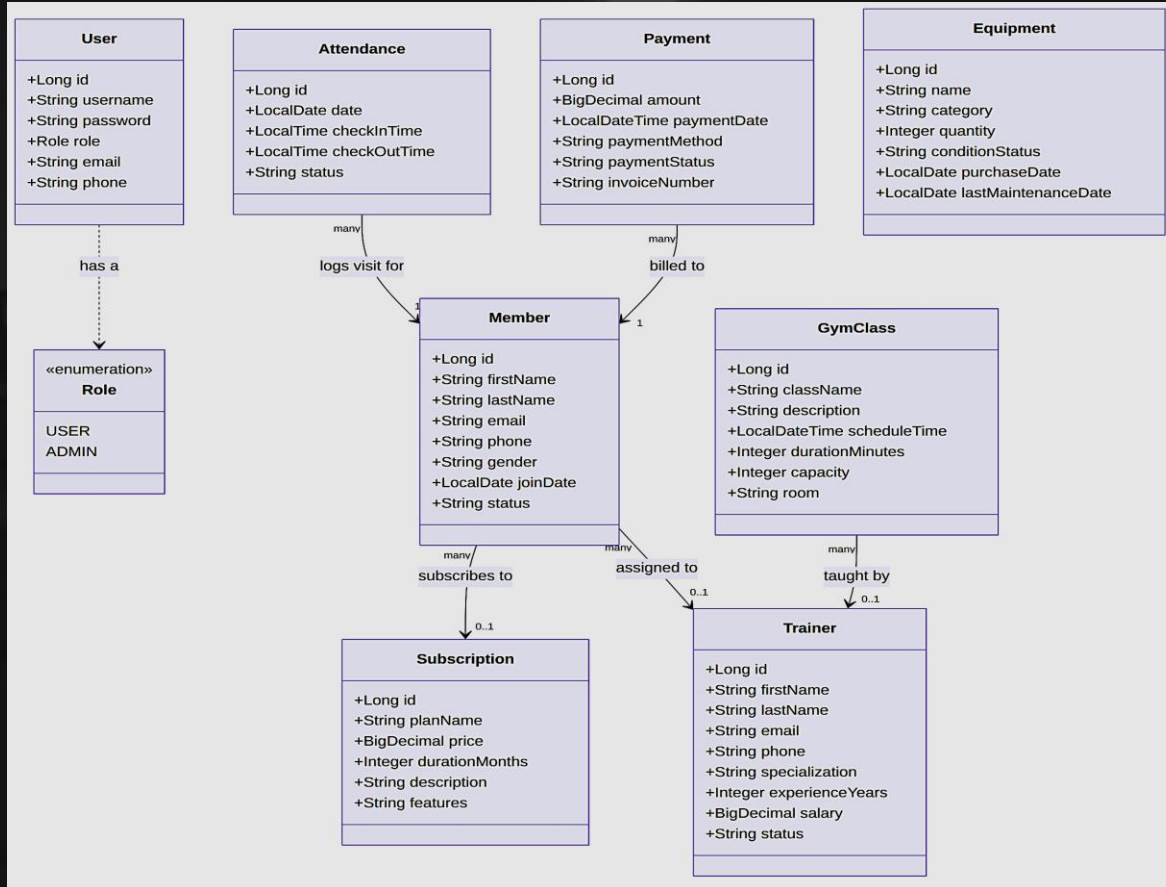
State & UI Refresh

Sub-second React state update and KPI chart refresh.

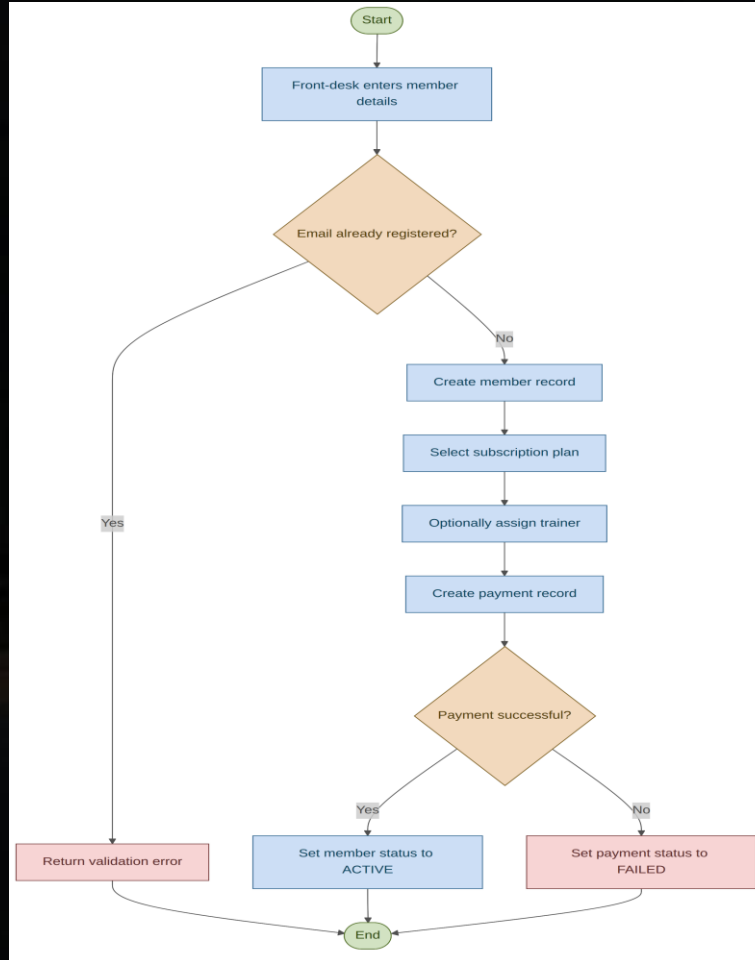
Use Case Diagram & System Boundaries



Class Diagram & Relational Entity Model



Activity Diagram



Admin Dashboard & Financial Analytics

Total Members

1,284

+12% this month

Active Trainers

18

Full capacity

Today Check-ins

142

Peak 5 PM - 8 PM

Monthly Revenue

\$48,250

98.5% collected

RECHARTS VISUAL ANALYTICS ENGINE

- Revenue Trend (Area Chart): Monthly subscription intake tracking across VIP Gold, Standard Pro, and Basic Student plans.
- Access Distribution (Bar Chart): Real-time hourly attendance volume telemetry showing peak facility usage.
- Dynamic Re-rendering: Dashboard components update instantly upon creating payments or scanning member check-in QR codes.

Interactive System Demonstration Workflow

DEMO STEP 01

Admin Login

Authorize session token.

DEMO STEP 02

Register Member

Add VIP Gold plan member.

DEMO STEP 03

Edit Trainer Roster

Update salary & assigned roster.

DEMO STEP 04

QR Check-in

Simulate barcode scan check-in.

DEMO STEP 05

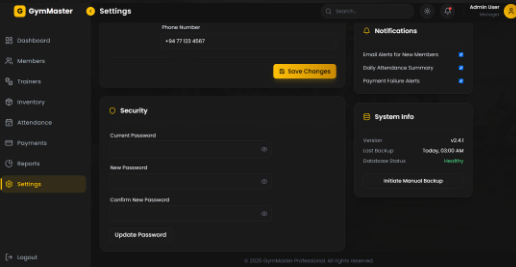
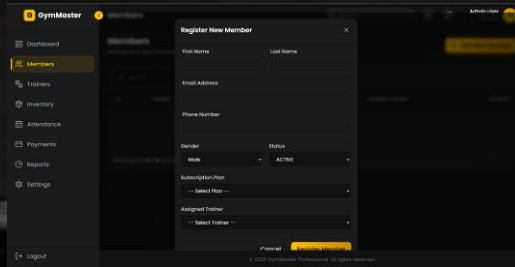
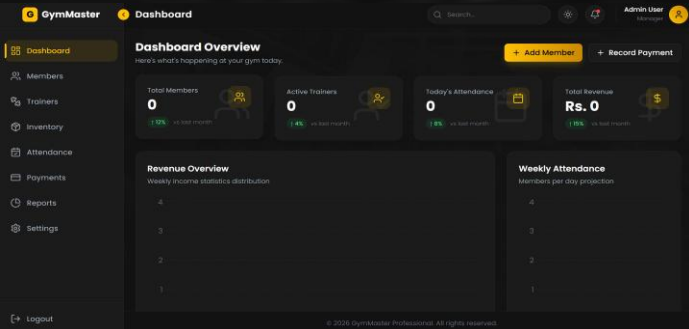
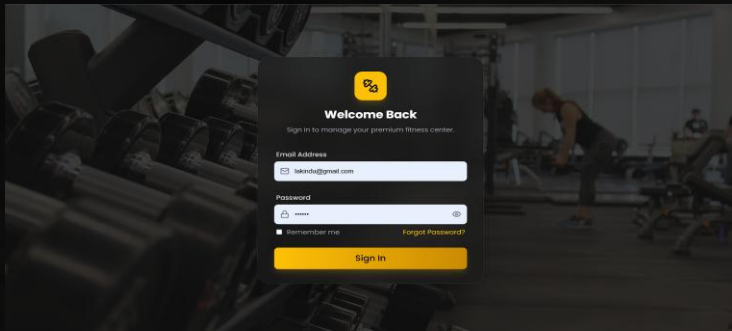
Record Payment

Generate plan renewal invoice.

DEMO STEP 06

Dashboard Refresh

Observe live KPI & chart updates.



GitHub Repository & Branching Strategy

PUBLIC GITHUB REPOSITORY LINK:

[pasinduliyagamage/Gym Management System: group project](https://github.com/pasinduliyagamage/Gym_Management_System)

01

Feature Branching

Isolate changes on feature/member-crud or feature/qrcode-scanner.

02

Pull Request

Submit PR against main with description and test screenshots.

03

Code Review

Peer verification and automated unit test build passes.

04

Main Deployment

Individual Team Contributions

Team Member	Assigned Module	Core Engineering Contribution
Lakshitha,sumiyuru	Members,class Management	Full CRUD, validation, search filtering & pagination logic.
Ashika	Trainers	Full CRUD, card grid view, trainerService.delete(id) execution.
Pasindu	Equipment Inventory	Full CRUD, category, quantity, condition badges & low stock alerts.
Sahan,Irushan ransara	Attendance,sudcription & Payments	Member check-in/out scanner, status verification, payment logs.
Pasindu	Dashboard & UI	Integrated system, dark luxury UI theme, Recharts engine.

The background image is a dark, atmospheric photograph of an industrial facility, possibly a steel mill or a large warehouse. The scene is dimly lit, with a strong red or orange glow emanating from the center, creating a sense of heat and intensity. Large, dark metal structures, including what appear to be cranes or support beams, are visible on both sides of a central walkway. The floor is dark and reflective, mirroring the light from the center. The overall mood is gritty and industrial.

CRUD Specifications

Member Management CRUD Specifications

Operation	REST Endpoint	Payload / Logic
CREATE	POST /api/members	Registers name, email, phone, VIP plan tier.
READ	GET /api/members	Fetches paginated list with real-time text filter.
UPDATE	PUT /api/members/{id}	Updates tier, status, contact info, or expiry.
DELETE	DELETE /api/members/{id}	Executes soft-deactivation or hard deletion.

Trainer Roster & Staff Management

Operation	REST Endpoint	Description & Fields
CREATE	POST /api/trainers	Registers trainer with specialization & monthly salary.
READ	GET /api/trainers	Displays card grid with assigned member counts.
UPDATE	PUT /api/trainers/{id}	Modifies salary, assigned members, or active status.
DELETE	DELETE /api/trainers/{id}	Removes instructor profile via <code>trainerService.delete(id)</code> .

Equipment Inventory & Attendance Tracking

EQUIPMENT INVENTORY

- Track Cardio, Strength, & Free Weights.
- Condition Flags: Optimal, Maintenance, Repair Due.
- Automated quantity & stock alert indicators.
- Service schedule logging & maintenance history.

ATTENDANCE & QR

- Sub-second contact-free QR check-in simulation.
- Real-time membership expiry status validation.
- Daily timestamp logs & peak access telemetry.
- Automatic badging: Valid, Warning, or Expired.

Equipment CRUD Specifications

Operation	REST endpoint	logic / required fields / sample payload
Create Equipment	POST /api/equipment	Saves a new Equipment record. Required: name, category, quantity, conditionStatus (GOOD, MAINTENANCE_REQUIRED, REPLACED), purchaseDate, lastMaintenanceDate.
Read All Equipment	GET /api/equipment	List of EquipmentDTO.
Read Equipment by Id	GET /api/equipment/{id}	Single EquipmentDTO.
Update Equipment	PUT /api/equipment/{id}	Updates any mutable fields (e.g., quantity, conditionStatus, dates).
Delete Equipment	DELETE /api/equipment/{id}	Deletes the row; 204 on success.

Attendance CRUD Specifications

Operation	REST endpoint	logic / required fields / sample payload
Create Attendance	POST /api/attendance	Inserts an attendance record with checkInTime. Required: memberId, date (ISO), checkInTime, status (PRESENT, ABSENT, LATE).
Read All Attendance	GET /api/attendance	Returns all attendance entries.
Read Attendance by Id	GET /api/attendance/{id}	Returns a single attendance entry.
Update Attendance	PUT /api/attendance/{id}	Sets checkOutTime and optionally updates status. Payload: { "checkOutTime": "18:30:00", "status": "PRESENT" }.
Delete Attendance	DELETE /api/attendance/{id}	Removes the record; 204 on success.

Payment CRUD Specifications

Operation	REST endpoint	logic / required fields / sample payload
Create Payment	POST /api/payments	Persists a payment linked to a member. Required: memberId, amount, paymentMethod (CARD, CASH, UPI, NETBANKING), paymentStatus (PAID, PENDING, FAILED), invoiceNumber, paymentDate (ISO-datetime).
Read All Payments	GET /api/payments	List of PaymentDTO.
Read Payment by Id	GET /api/payments/{id}	Single payment details.
Update Payment	PUT /api/payments/{id}	Allows status change or correction of amount/method. Payload may contain any subset, e.g. { "paymentStatus":"FAILED" }.
Delete Payment	DELETE /api/payments/{id}	Removes the payment; 204 on success.

Gym Class - CRUD

Operation	HTTP Method & URL	Description
Create	POST /api/classes	Creates a new gym class. The request body must be a GymClassDTO JSON payload. Returns the created object with 201 Created .
Read (all)	GET /api/classes	Retrieves a list of all gym classes. Returns a JSON array of GymClassDTO.
Read (single)	GET /api/classes/{id}	Retrieves a single class by its database id. Returns a GymClassDTO.
Update	PUT /api/classes/{id}	Updates an existing class (all mutable fields). The request body is a GymClassDTO. Returns the updated GymClassDTO.
Delete	DELETE /api/classes/{id}	Removes a class by id. Returns 204 No Content on success.

Subscription CRUD Specifications

Operation	REST endpoint	logic / required fields / sample payload
Create Subscription	POST /api/subscriptions	Adds a new membership plan. Required: planName, price, duration (e.g. "12 months"), status.
Read All Subscriptions	GET /api/subscriptions	Returns all plans.
Read Subscription by Id	GET /api/subscriptions/{id}	Single plan data.
Update Subscription	PUT /api/subscriptions/{id}	Updates any plan field (price, status, etc.).
Delete Subscription	DELETE /api/subscriptions/{id}	Deletes the plan; 204 on success.

Conclusion & Future Engineering Roadmap

KEY ACHIEVEMENTS DELIVERED

- Delivered centralized CRUD for Members, Staff, and Assets.
- Real-time contact-free QR attendance check-in scanner.
- Layered separation (UI -> REST API -> JPA -> H2).
- Responsive Dark Luxury UI with Recharts KPI engine.

FUTURE ROADMAP ENHANCEMENTS

- Role-Based Access Control (RBAC) with granular JWT.
- React Native Mobile App for member workout tracking.
- online billing.
- Automated SMS & Email notifications

Defense Q&A Session

THANK YOU FOR YOUR TIME

Questions?

We welcome questions, feedback, and technical inquiries from the panel.